

# **PROJECT 05 REPORT**

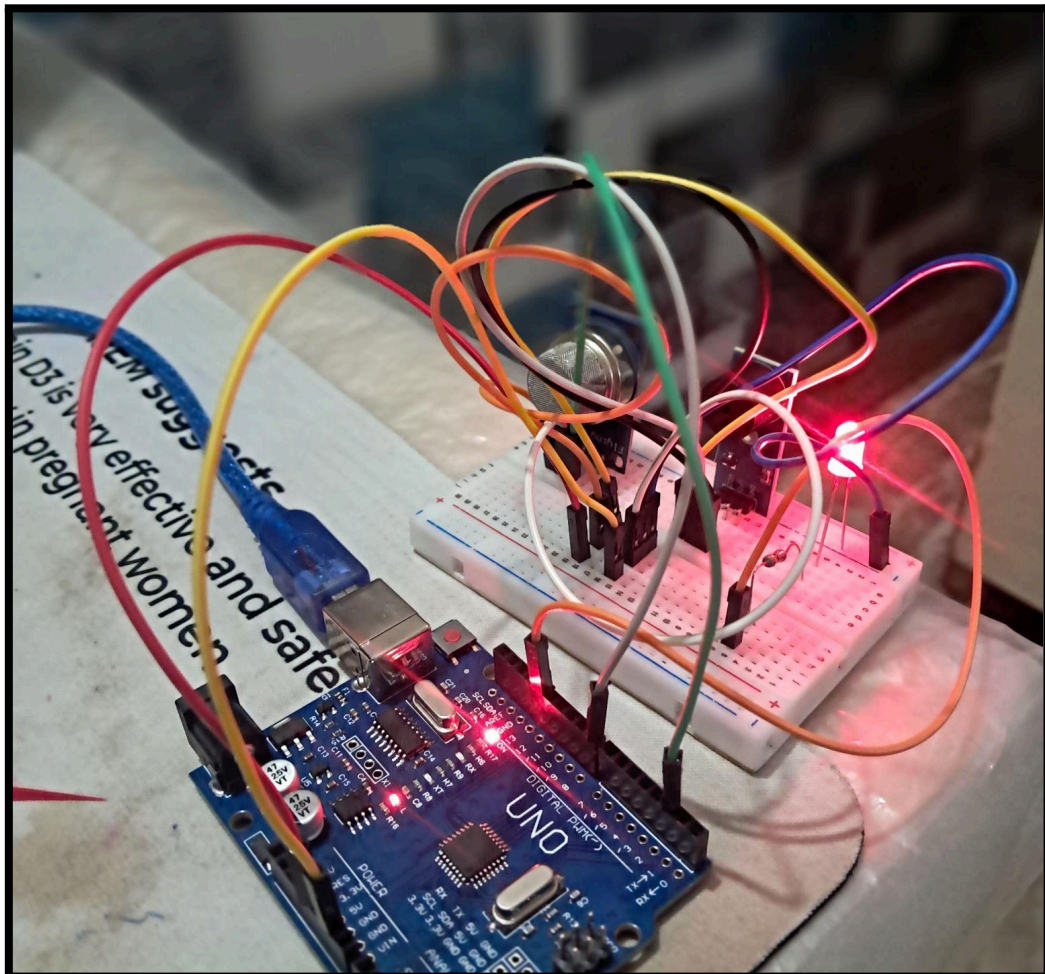
## **Gas Leakage Detection System using MQ2 Sensor**

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**Duration :** 2 days



# 1. Objective

The objective of this project was to develop a gas leakage detection and alert system using the MQ2 gas sensor and Arduino Uno.

The system was designed to continuously monitor the surrounding air for the presence of smoke and combustible gas vapors and generates an alarm whenever the detected concentration exceeds a predefined threshold.

# 2. Components Used

| Component             | Quantity |
|-----------------------|----------|
| Arduino Uno R3        | 1        |
| Breadboard            | 1        |
| MQ2 Gas Sensor Module | 1        |
| Active Buzzer Module  | 1        |
| LEDs (Red)            | 1        |
| Resistors(300Ω)       | 1        |
| Jumper Wires          | Multiple |
| USB Cable             | 1        |

### 3. Software Used

- Arduino IDE
- Serial Monitor
- Windows Laptop

### 4. Theory

The MQ2 is a gas and smoke detection sensor capable of detecting combustible gases and vapors such as LPG, propane, butane, hydrogen, smoke and chemical vapors

The sensor contains a sensing element and an internal heating element which continuously heats the sensing surface.

The MQ2 module provides two outputs:

- Analog Output (AO)
- Digital Output (DO)

#### **Analog Output (AO)**

Provides continuously varying values according to gas concentration.

The analog values can be read using: **analogRead(A0);**

## Digital Output (DO)

The module also contains a comparator circuit that compares the sensor value against a threshold set using the onboard potentiometer.

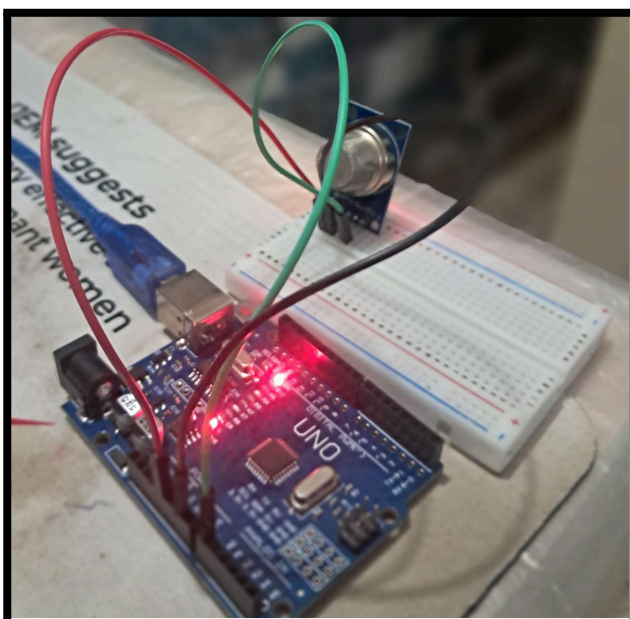
The digital output can be read using: **digitalRead(2);**

## 5. Tasks Performed

### ➤ Understanding MQ2 Sensor Module

The MQ2 sensor module was studied to understand its operating principle, pin configuration, onboard potentiometer, power indicator LED, output indicator LED, analog output (AO) and digital output (DO).

### ➤ Reading Analog Sensor Values



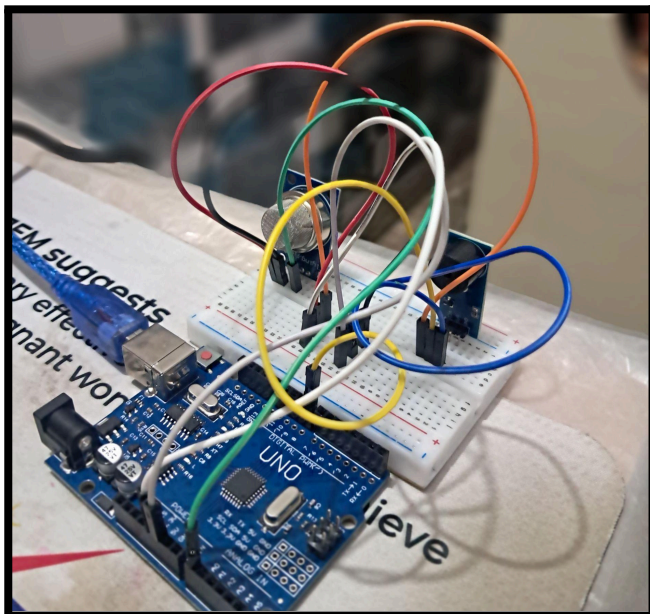
**Video**

The analog output of the MQ2 sensor was connected to Arduino and continuously monitored using the Serial Monitor.

The sensor was first allowed to warm up and stabilize. Readings were then recorded under normal room conditions and under smoke exposure.

The sensor values increased significantly when exposed to smoke and gradually returned to normal after the smoke source was removed.

## ➤ AO Based Gas Leakage Alarm



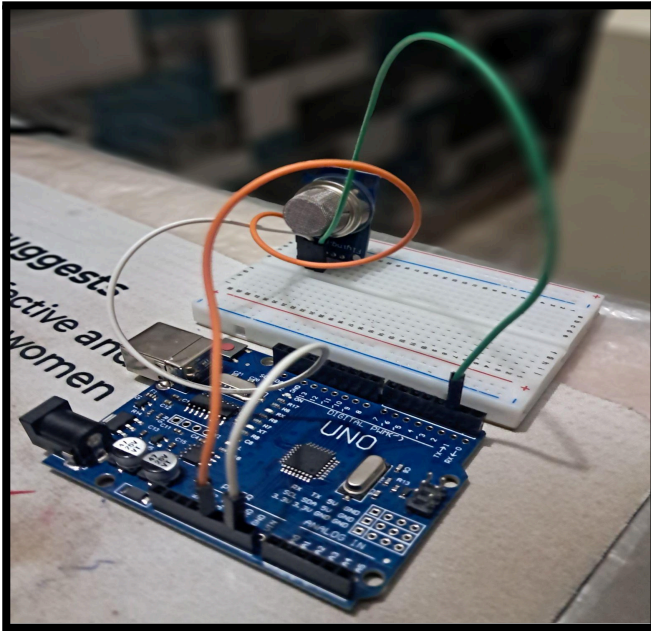
**Video**

A gas leakage alarm system was implemented using the analog output of the MQ2 sensor. The sensor value was continuously compared to a threshold value.

The active buzzer generated an alarm when smoke concentration exceeded the selected threshold.



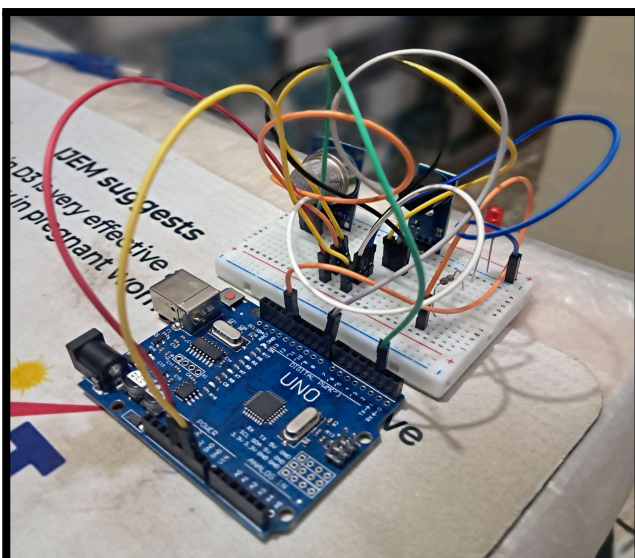
## ➤ Understanding Digital Output (DO)



**Video**

The digital output pin of the MQ2 sensor was studied and interfaced with Arduino and the onboard potentiometer was adjusted to observe how the comparator threshold affected the digital output.

## ➤ DO Based Gas Leakage Alarm



**Video**

The final system was implemented using the MQ2 digital output, LED indicator and active buzzer.

The system functioned as a complete gas leakage alarm and safety monitoring device.

## 6. Learning Outcomes

During this project, the following concepts were learned:

- MQ2 Gas Sensor Operation
- Gas and Smoke Detection
- Analog Output (AO)
- Digital Output (DO)
- Comparator Circuits
- Potentiometer Based Threshold Adjustment
- Threshold Based Alarm Systems
- Active Low Output Devices
- Sensor Response Time
- Sensor Recovery Time
- Embedded System Debugging
- Safety Monitoring Systems

## 7. Problems Faced

- I. The MQ2 sensor readings continuously changed after power up. **(The sensor requires warm up time for the internal heating element to stabilize and was allowed to operate for a short period before recording observations)**

- II. The buzzer behaved opposite to the expected logic.**(The active buzzer module used active low logic)**
- III. The gas alarm remained active even after removing the smoke source.**(The sensor was given sufficient recovery time to return to normal operating conditions)**

## 8. Observations

- MQ2 readings increased significantly in the presence of smoke and vapors.
- Sensor values required warm up time before stabilizing.
- Detection was not instantaneous and required several seconds to respond.
- Recovery time was longer than detection time.
- The onboard potentiometer directly controls the digital output threshold.
- The output indicator LED reflects the state of the comparator output.
- The sensor successfully detected incense stick smoke and perfume vapors.
- Active low devices require careful testing before implementation.



## **9. Applications and Future Scope**

Gas leakage detection systems are widely used in homes, industries, laboratories, kitchens and storage facilities to improve safety by detecting smoke and combustible gas leaks at an early stage.

This project can be further improved by additional safety features such as automatic exhaust fan control, cloud monitoring, and multi gas detection capabilities can also be incorporated to create a more advanced monitoring system.

## **10. Conclusion**

This project successfully demonstrated the development of a complete gas leakage detection and alarm system using the MQ2 sensor and Arduino Uno.

The successful detection of smoke and chemical vapors demonstrated the effectiveness of the system in practical safety monitoring applications.